Checkpoint 5 — k-means Objective & PCA — Student Version

INFO 521 — Machine Learning Foundations · checkpoint (prompts & criteria; graded in class)

> **i How this works**
>
> This checkpoint is sat **in class, closed-book** (~15–20 minutes, one derivation) and graded Satisfactory / Not-Yet against the criteria below. Version B is the alternate-version retake, offered the following week; you keep your best result. Knowing the prompt in advance is the point — practice until you can reproduce the derivation unaided.

Sitting: Week 6 · retake: Week 7 (PCA version) · gates Project 2 · Gate B.

Gates: Project 2.3 (Clustering) — and with it, Gate B. **Format:** in-class, closed-book, ~15–20 min, one derivation/explanation. Graded Satisfactory / Not-Yet. **Retakeable** (an alternate version is provided below).

This checkpoint carries the unsupervised-gate recall outcomes: the k-means objective and why model selection for K is non-trivial (Version A), and the PCA procedure and the meaning of its eigenstructure (Version B).

Prompt (Version A)

State the **k-means objective** J as a function of the cluster assignments and centroids. Explain why choosing K by **minimizing J over K** trivially fails, and describe **one principled method** for selecting K.

Prompt (Version B — retake)

Describe the **PCA procedure** step by step for a centered data matrix, and state what the **eigenvalues** and **eigenvectors** of the covariance matrix represent.

Satisfactory criteria

- (A) Correct objective J (within-cluster sum of squared distances); states that J is **non-increasing in K** and hits 0 at $K = N$, so minimizing J always prefers more clusters; names a valid criterion (elbow, silhouette, or gap) and describes it.
- (B) Lists center → covariance → eigendecompose (or SVD) → sort → project; states eigenvectors = principal directions (orthogonal axes of maximal variance), eigenvalues = variance along each, and the ratio $\lambda_i / \sum_j \lambda_j$ = proportion of variance explained.